ANALYSIS REPORT

CVP 1.2 Curve Payload Diagnostics Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates a screened `TE Curve Verification Pipeline` candidate set with full truth/prediction curve payloads. It does not train models, alter the dataset structure, or provide future curve samples to runtime model inputs.

- Run Instance: `2026-05-28-19-55-32__track2c_curve_payload_diagnostics`
- Config Path: `config\paper_reimplementation\rcim_ml_compensation\reference_family_vs_feedforward\full_track2_matrix_template.yaml`
- Output Directory: `output\validation_checks\track2_curve_payload_diagnostics\2026-05-28-19-55-32__track2c_curve_payload_diagnostics`
- Evaluated Curve Payload Count: `1164`
- Harmonic Orders: `1, 2, 3, 4, 5, 6, 8, 10, 12, 19`
- Stored payload samples are downsampled for repository-size control.

Method

Diagnostics are computed on full held-out `TE` curves after each candidate produces pointwise predictions through its normal causal input path. The report separates the validation surface from the runtime input contract.

Computed diagnostics

- peak-to-peak amplitude error and residual peak-to-peak ratio;
- selected-harmonic amplitude and wrapped phase error;
- local derivative `RMSE` ;
- residual second-derivative smoothness;
- residual lag-one autocorrelation;
- per-revolution closure mismatch and deterministic stitched-boundary surrogate.

Candidate Diagnostic Ranking

Rank	Candidate	Surface	Directions	Mean MPE [%]	Harm. Amp Err. [%]	Harm. Phase Err. [deg]	P2P Err. [%]	Diag. Score
1	`rcim_retuned_GBM19_Fw`	Fw	forward	2.371752	33.021378	28.258963	8.089395	7.227298
2	`periodic_gru_sequence_Bw`	Bw	backward	5.465990	14.792262	8.570493	5.458720	7.519285
3	`periodic_lstm_sequence_global`	global	backward, forward	6.119950	17.757138	11.622748	5.863664	8.621953
4	`tree_global`	global	backward, forward	6.854374	19.143975	11.526651	18.028018	9.503512
5	`tree_Bw`	Bw	backward	7.051484	19.173838	7.988429	17.916583	9.530644
6	`harmonic_regression_Bw`	Bw	backward	8.170944	9.758297	5.090618	18.926302	9.558129
7	`residual_harmonic_lstm_sequence_sparse_rcim_Bw`	Bw	backward	7.510266	27.574565	7.534689	10.580972	10.800185
8	`residual_harmonic_lstm_sequence_sparse_rcim_global`	global	backward, forward	7.409051	31.641801	28.653671	13.339015	12.157611
9	`rcim_retuned_GBM19_Bw`	Bw	backward	5.398275	32.964035	76.738397	5.703129	12.705128

Interpretation

The strongest screened diagnostic score is `rcim_retuned_GBM19_Fw`. The score is an analysis aid, not a registry-promotion rule.

The strongest screened forward candidate is still the paper-reference `rcim_retuned_GBM19_Fw`. The forward repository-owned branch therefore remains open and should not be closed just because the backward neural branch is stronger.

The strongest screened repository-owned backward candidate is `periodic_gru_sequence_Bw`. It is close to `rcim_retuned_GBM19_Bw` on mean percentage error, but has much better selected-harmonic phase behavior in this diagnostic pass.

`periodic_lstm_sequence_global` is the strongest screened global-surface neural candidate. It keeps low peak-to-peak error on both directions and must remain a separate global branch, not a weaker substitute for the backward-only `periodic_gru_sequence_Bw` branch.

`harmonic_regression_Bw` has the cleanest selected-harmonic amplitude and phase diagnostics in the backward-only set, but its mean percentage error and peak-to-peak error remain worse than the periodic temporal candidates.

`tree_Bw` and `tree_global` confirm the visual concern from TE Curve Verification Pipeline: scalar error can look competitive while peak-to-peak and shape diagnostics remain weaker.

Paper-reference and tree-bank candidates may remain strong on some curve metrics while still being less attractive for deployment. Repository-owned neural candidates should therefore be judged by both diagnostic behavior and future export/runtime feasibility.

Decision

The next work should not start from `tree`, despite its scalar strength. The practical direction is a set of parallel curve-aware retraining or reranking branches: keep searching the `Fw` surface for a deployable repository-owned candidate, prioritize `periodic_gru_sequence_Bw` on the `Bw` surface, and carry `periodic_lstm_sequence_global` as the dedicated global-surface candidate. Harmonic/phase-aware validation stays separate from runtime inputs.

Runtime Input Boundary

All diagnostics are computed after prediction. Candidate models still consume only current point-level operating state, supported short causal history, or derived causal features. Full curves remain a validation and promotion surface only.

Machine-Readable Artifacts

- `output\validation_checks\track2_curve_payload_diagnostics\2026-05-28-19-55-32__track2c_curve_payload_diagnostics\candidate_payload_diagnostics.csv`
- `output\validation_checks\track2_curve_payload_diagnostics\2026-05-28-19-55-32__track2c_curve_payload_diagnostics\curve_payload_diagnostics.csv`
- `output\validation_checks\track2_curve_payload_diagnostics\2026-05-28-19-55-32__track2c_curve_payload_diagnostics\curve_payload_samples.jsonl`
- `output\validation_checks\track2_curve_payload_diagnostics\2026-05-28-19-55-32__track2c_curve_payload_diagnostics\track2_curve_payload_diagnostics_summary.yaml`